Hackathon Project Ideas
Targeted Real-Life Solutions

Hacktoberfest 2026 - 3 Prize Categories

1. Best Use of Gemma 4 (DeepMind Google)

2. Best Open-Source AI Project (DigitalOcean)

3. Best Use of Adaption AI (Event Prize)

Prepared: October 16, 2026

Focus: Buildable in 24-48h, high social impact, eligible for multiple categories.

1. Prize Categories Overview

A. Best Use of Gemma 4 - DeepMind Google

Lightweight, open model with multimodal intelligence via Gemini API. Gemma 4 family: E2B/E4B (mobile/IoT edge), 12B (laptop-ready, native audio + vision, encoder-free), 26B MoE A4B and 31B (workstation). Features: text + image + audio input, text output, 256K context, 140+ languages, thinking mode for reasoning, agentic workflows, Apache 2.0 license. Runs on 16GB RAM laptops, phones. Ideal for offline, privacy-preserving, low-resource apps.

B. Best Open-Source AI Project - DigitalOcean

Build an original project that uses open-source or open-weight AI as an important part of how it works. 2026 theme: 'AI belongs to everyone'. Expected artifacts: skills.md, open-source agent, fine-tune an open-weight model. Deploy on DigitalOcean Gradient (GPU), App Platform, Droplets, Storage. Rewards: Exclusive Hacktoberfest 2026 Winner DEV Badge + MLH Swag Bag.

C. Best Use of Adaption AI - Event Prize

Build innovative AI-powered solution using Adaption AI. Platform: Adaptive Data + AutoScientist. Adaptive Data: optimize, expand, invent training datasets across 242 languages, from CSV/JSON/PDF/DOCX/HuggingFace/Kaggle. Forge turns unstructured docs into training data. Blueprint controls length/tone/safety. AutoScientist co-optimizes data + training recipe automatically. SDK: pip install adaption. 82% average quality gains reported.

2. Winning Strategy: Triple-Dip

Do NOT build 3 separate projects. Build ONE project eligible for all 3:

1. Core = Gemma 4 12B via Gemini API + Ollama local fallback (satisfies Gemma 4)
2. Open-source = MIT repo with skills.md agent + HuggingFace dataset/model + DigitalOcean Gradient deploy (satisfies DigitalOcean)
3. Customization = Use Adaption Adaptive Data to invent/adapt dataset for local language/domain, show before/after win-rate, loss, quality (satisfies Adaption AI)

Demo script: Show offline mode + local language + Adaption quality comparison.

3. Six Project Ideas - Detailed

Project 1: ClinicOffline - Rural Health Triage [Best for Gemma 4]

- **Problem:** Clinics with no internet / no doctor cannot interpret prescriptions, rashes, lab slips.
- **Build:** Mobile-first PWA running Gemma 4 E4B/12B on-device. Input: photo of skin condition / prescription / lab slip + voice symptom. Output: triage level, referral urgency, plain-language explanation in local language. Thinking mode for step-by-step reasoning. No data leaves phone.
- Offline + privacy: runs on cheap Android, 16GB RAM
- Multimodal: vision + audio natively

- Multilingual: 140+ languages
- Stack: Gemma 4 E4B, LiteRT / Ollama, React PWA, Gemini API
- **Why it wins:** Gemma edge story judges love. High impact.

Project 2: AgriLingua - Crop Doctor for Smallholder Farmers [Gemma 4 + Adaption]

- **Problem:** Extension workers scarce. Crop disease advice not available in Hausa, Swahili, Yoruba, Hindi.
- **Build:** Farmer snaps diseased leaf + 10s voice note. Gemma 4 12B vision + audio -> diagnosis + organic/chemical treatment in local language + audio playback. Use Adaption Invent + Expand to generate data where none exists.
- Adaption Adaptive Data: invent dataset for local crops, translate/localize
- Gemma 4 12B: unified encoder-free vision/audio
- Open-source skill: crop-doctor skills.md
- Deploy: DigitalOcean App Platform + Storage for images
- **Why it wins:** Perfect Gemma + Adaption combo. Very strong live demo.

Project 3: LearnInMyTongue - Textbook to Local Lessons [Best for Adaption AI]

- **Problem:** Students fail due to language, not intelligence. Textbooks stuck as English PDFs.
- **Build:** Upload PDF/DOCX/PPTX -> Adaption Forge extracts/structures -> Blueprint sets tone/length/safety -> instruction dataset + preference pairs -> fine-tune via Together Fine-Tuning / AutoScientist OR prompt Gemma 4 256K context for quizzes, summaries, audio lessons.
- Literal Adaption use case: unstructured docs to training data
- 256K context: ingest whole textbook
- Multilingual: 242 languages (Adaption) + 140 (Gemma)
- Export: HuggingFace dataset + model
- **Why it wins:** Strongest fit for Adaption prize.

Project 4: ShopBooks - Receipt-to-Books for SMEs [Best for DigitalOcean]

- **Problem:** Small shops use paper, cannot track profit, tax, or qualify for loans.
- **Build:** Photo of receipts/invoices/handwritten ledger -> Gemma 4 doc understanding (OmniDocBench SOTA) -> sales table, profit, low-stock alert, WhatsApp-style summary. Open agent with bookkeeping skills.md.
- Gemma multimodal doc parsing
- DigitalOcean Gradient + Postgres + App Platform
- Open-source agent + skills.md
- Future: Adaption Blueprint for financial tone/safety
- **Why it wins:** Business value clear. Easy MVP in hackathon.

Project 5: CivicCheck - Misinformation Checker [Best for Open-Source]

- **Problem:** Viral WhatsApp screenshots spread health/election rumors.
- **Build:** Paste screenshot + claim -> Gemma 4 fact-check agent explains verdict simply, with sources. Publish reusable skills.md + dataset on HuggingFace. Host on DigitalOcean.
- 2026 theme match: 'write your first skills.md'

- Fully open-source repo, model, data
- Agent + function calling (Tau2 benchmark strength)
- Low compute, high demo impact
- **Why it wins:** What MLH/DEV want this year.

Project 6: HearMe - Accessibility Captions

- **Problem:** Deaf / low-literacy users locked out of audio/visual content.
- **Build:** Gemma 4 E2B on-device: live audio -> captions + simplified visuals; image -> spoken local-language description. Works offline, no cloud cost.
- Edge AI: E2B maximum efficiency
- Native audio input (E2B/E4B/12B)
- Social impact + inclusion story
- Stack: Gemma Edge Gallery, LiteRT, Android
- **Why it wins:** Impact prize potential.

4. Recommended MVP Scope (24-48h)

Pick ONE idea (AgriLingua or LearnInMyTongue recommended). MVP must include:

Day 1: Gemma 4 chat + image/audio upload working via Gemini API, basic UI, deploy to DigitalOcean App Platform
Day 2: Add skills.md open-source agent file, run ONE Adaption Adaptive Data job (import 50-100 rows -> adapt -> export), show before/after quality screenshot, record 2-min demo in local language + offline mode

Must-haves for judging: live link, GitHub README with architecture diagram, DEV post, 90-sec video, Adaption run ID/screenshot, DO deploy URL.

5. Tech Checklist

Gemma 4: Google AI Studio / Gemini API, Ollama (gemma3 / gemma4), LM Studio, HuggingFace weights, Kaggle
Adaption: pip install adaption, ADAPTION_API_KEY, Adaptive Data quickstart, AutoScientist quickstart
DigitalOcean: Gradient GPU, App Platform, Droplets, Spaces Storage, Postgres
Common: Python FastAPI + React/Vite PWA, Tailwind, HuggingFace Hub, DEV Challenge post

Next: Choose 1 idea, confirm team size, stack (Python/React?), target language/domain, and I will generate architecture + task list + starter code.